An Audit of a TF-IDF Ensemble: Evaluation Validity, Inference Reconstruction, and Accounting Boundaries in Harmful-Content Classification

Abstract

We audit a stacked TF-IDF ensemble for harmful-content classification and trace how evaluation and parameter accounting relate to the predictor actually used. The recorded three-fold run contains 47,817 labeled training rows, 11,955 test rows, 230,105 TF-IDF features, six base candidates, and a logistic regression stacker. The selected weighted ensemble records pooled out-of-fold macro-F1 0.5314, compared with 0.5221 for the strongest base candidate. We distinguish label-free representation fitting, aggregate class statistics, supervised meta-feature dependencies, stopping, and selection. Regenerated seed-42 folds place 308 exact-text groups across folds; 52 repeated-text groups contain conflicting labels. Reconstructed ensemble weights range approximately from 0.111846 to 0.158125. Their observed advantage over uniform averaging is 0.0008, not a bound on selection bias. The separately profiled identity module is absent from the prediction path. Six base logistic regression fits imply 4,141,908 coefficients and intercepts, while the exported value 0.183 is derived from a target ratio. Source observations, retained outputs, data checks, and arithmetic are labeled separately. A paired LR comparison yields macro-F1 0.522061 with global TF-IDF and 0.518872 with fold-fitted TF-IDF, holding class weights fixed. The audit supplies controlled rerun protocols. It introduces no new classifier or measured deployment advantage.

Keywords: Harmful-content classification, Evaluation validity, Stacked generalization, Accounting boundaries, Reproducibility.

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1. Introduction

A harmful content classifier's reported accuracy and efficiency are interpretable only when they refer to a specified prediction procedure. In sparse ensembles, predictions can depend on fitted representations, calibrated base estimators, and a second-stage head. Counting the head alone does not describe the system that produces its inputs [1].

We distinguish three objects: the *inference object*, which maps raw text to predictions; the *evaluation object*, which includes fitting, validation, and selection; and the *accounting object*, which defines the artifacts and computation included in a resource claim. Stacking [2] makes their relationship consequential: excluding a row from its own base fit does not necessarily isolate the full procedure when meta training features depend on meta validation labels. Global preprocessing and repeated selection create additional dependencies [3, 4, 5].

We examine one recorded implementation from a public datathon. The updated notebook, retained outputs, competition description, and raw CSVs permit source inspection, data checks, inference reconstruction, and arithmetic accounting. Historical row-level OOF predictions and fitted artifacts remain unavailable. We execute new data checks and a controlled LR comparison; these are distinct from replaying the original predictions. Documentation describes a mixed Bangla, English, and Banglish corpus; preprocessing alone would not establish this scope.

The contribution is an inspectable audit linking the three objects within one artifact, followed by a corrected prospective protocol. It is not a new algorithm, a demonstration of predictive superiority, or a measured deployment advantage. Section 3 defines the evidence categories and verification limits used throughout.

2. Related Work

Hate speech, offensive language and implicit harm require documented annotation decisions [6]. Functional tests probe negation and counter speech [7]; harm and bias analysis requires stating whose interests are considered [8]. BD-SHS [9] and BanglaBERT [10] provide Bangla context, not directly comparable scores or verified source datasets for this corpus. Data statements help document population and curation scope [11].

Stacked generalization learns from base predictions [2]. Super Learner studies cross-validated combination and risk minimization [12]; the source's squared-F1 heuristic is different. Hastie et al., Section 7.10.2 [13], explain why model assessment must include preparation and selection within permitted training data. Their supervised-selection example is not a numerical estimate of label-free TF-IDF bias here. Selection bias [3, 4] and leakage [5] motivate tracing full feature ancestry. Calibration adds fitted state [14] and requires separate reliability diagnostics [15]. Overlapping folds complicate uncertainty [16]; paired tests require retained predictions and an appropriate design [17].

Green AI [1], energy reporting [18], reproducibility [19], and model cards [20] distinguish computation, replay and intended use. This study applies those established principles to one prediction and accounting record.

3. Study Design, Evidence, and Dataset Scope

3.1. Design and Audit Method

The unit of analysis is the relationship among the implementation, retained outputs, and supported claims. We trace invoked fitting, prediction, selection, profiling, and saving operations through their inputs; comments do not establish execution. We inspect the supplied updated notebook and check raw CSV shapes, labels, identifiers, and exact text duplication. We also reconstruct metrics from its embedded confusion figure. These checks substantiate source structure and internal consistency, not authenticated historical execution.

Each claim has an evidence type: S (source observation), R (retained historical output or explicitly identified new execution), D (direct CSV check), A (arithmetic reconstruction), X (unverified external quantity), or P (prospective proposal). An unperformed experiment is stated as such. New checks regenerate fold indices and inspect duplicates, export structure, and weight arithmetic. The LR comparison uses a new software environment; the six-candidate historical pipeline and nested stack are not replayed. Source locations, full file digests, regenerated folds, and execution status are supplied in the artifact ledger and supplement. The availability statement identifies the review copy.

3.2. Available Artifacts and Version Boundaries

The canonical notebook has one code cell, 971 source lines, execution count one, two output streams, and one figure (S). Metadata records execution on 20 April 2026 for 9,142.89 seconds (R). Python metadata lists 3.12.12; historical package versions are absent. Output reports `Device: cuda`, while notebook metadata disables GPU. These retained fields are not independently authenticated execution records.

The author now reports runtime 2 h 32 min 37 s on GPU T4 ×2, public score 3.78469, and private score 3.78930 (X). These were supplied in correspondence, without a linked scoring receipt or execution record. The reported duration is 9,157 seconds, approximately 14.11 seconds longer than notebook metadata (A). GPU allocation does not show that the classical estimators used GPU acceleration; the source sets no GPU device for LightGBM. The platform scores are neither macro-F1 measurements nor evidence resolving the missing scorer.

Table 1: Directly checked supports (D); proportions and globally balanced class weights (A). The source computes weights before splitting (S).

Class	Support	Proportion (%)	Weight
Explicit	23,756	49.681	0.670946
Subtle	18,163	37.984	0.877553
Neutral	5,898	12.335	2.702442
Total	47,817	100.000	

We use only the three-class 2026 archive. Historical variants are separated in Supplement S1. They do not support claims about this selected path.

3.3. Task and Checked Corpus

The supplied competition description maps 0 to Explicitly Toxic/Hateful, 1 to Subtly Toxic, and 2 to Non-toxic/Neutral. Training columns are `id`, `text`, and `y`; test columns are `id` and `text`. Shapes are (47,817, 3) and (11,955, 2) before processed columns are added (D), totaling 59,772 rows. Table 1 reports the class supports, which also match the updated notebook's printed counts.

The source uses $w_c = n/(3n_c)$ for LR, SVC, SGD, Meta-LR, and LightGBM sample weights. CNB has no explicit class weight argument. Since weights depend on every training label, they also depend on each fold subsequently used for assessment. The test rows have no labels and support no accuracy measurement.

Exact raw-string checks find 621 repeated training rows and 54 repeated test rows beyond first occurrences, and 254 distinct texts shared between train and test (D). New regenerated-fold counts are reported in Section 6.5. Identifiers are unique within each three-class file and disjoint between its train and test sets. Text overlap and identifier uniqueness are different properties. We neither estimate the effect of duplication nor perform a near-duplicate, shared-author, or thread analysis.

3.4. Documentation and Population Scope

The supplied description calls the corpus manually annotated Bangla, English and Banglish social-media content, emphasizing subtle toxicity, code mixing and noise. Language proportions and annotation agreement are not measured. Its details document lists CC BY-NC-ND 4.0, but supplies no full annotation manual, consent records or annotator-welfare documentation. Label names and script cues do not validate sarcasm detection or language identification. We describe the supplied documentation, not whether the original release had additional records.

4. Audited Prediction Pipeline

4.1. Cleaning, Signals, and Numerical Features

Let x be the original string after missing-value handling. The combined processed text is

$$s(x) = C(x) \,\|\, \texttt{SSEP} \,\|\, H(x), \tag{1}$$

where C cleans text and H produces signal tokens from the original input. The cleaner applies NFKC normalization, lowercasing, URL and mention substitutions, hashtag-marker removal, emoji substitution, repeated-letter reduction, repeated-punctuation markers, and whitespace compression (S). Signals retain access to original capitalization and punctuation rather than only the cleaned string.

The explicit lexical union contains 21 Bangla, 23 transliterated, and 29 English entries, totaling 73 distinct strings; the subtle inventory contains 25 entries (S). Lexical detection tests substring presence, counting each distinct entry at most once per text, rather than counting every occurrence. Signals include capped repetition of explicit, subtle, sarcasm, threat, identity-related, script, capitalization, punctuation, length, and neutral markers. The string "this is a salad" activates `sala`, a transliterated term used abusively in Bangla and Hindi; it also has benign and kinship uses. Here it is a substring inside "salad", not an abusive utterance. The match records a lexical condition rather than intent.

Twenty numerical features are generated without fitted scaling. They cover length, word count, lexical diversity, punctuation, emoji runs, Bangla regex runs, script indicators, capitalization, lexical hits, pattern flags, a combined cue count, and short/long indicators. One feature divides the number of Bangla regex runs by character length; it is not a measured fraction of Bangla characters. Numerical features are supplied to LightGBM only. A marker named for hate or neutrality is a code identifier, not a validated social construct.

4.2. Four Globally Fitted Sparse Branches

Table 2 lists the feature branches and recorded widths. All use sublinear term frequency and are fitted on all processed training texts before the three-fold split. Test texts are transformed, not fitted on (S). Under the documented default interpretation of omitted smoothed-IDF and normalization options, a branch computes

$$\begin{aligned} u_{ij} &= (1 + \ln t_{ij}) \left[1 + \ln \frac{n+1}{\mathrm{df}_j + 1}\right], \quad t_{ij} > 0, \\ \phi(x_i) &= u_i / \|u_i\|_2, \end{aligned} \tag{2}$$

with $u_{ij} = 0$ otherwise. Historical package versions and fitted arrays are absent; current documentation is not evidence of the original runtime.[1]

CNB receives both character branches, totaling 130,000 columns. LR, SVC, and SGD receive the 230,105-column TF-IDF matrix. LightGBM receives that matrix plus 20 numerical columns. The global map ϕ exposes validation covariates to vocabulary construction, document-frequency filtering, feature caps, and IDF. A later estimator fit that excludes a row cannot undo this earlier representation fitting.

4.3. Base Candidates and Stacker

The source forms shuffled stratified three-fold splits with seed 42. Six base candidates are trained: two LR configurations, calibrated SVC, calibrated CNB, calibrated SGD, and LightGBM [21]. Table 3 gives explicit settings. The two LR candidates share an estimator family; six candidates do not imply six distinct algorithmic families.

Calibration adds fitted state to the probability path [14]. SVC and SGD use global class weights, whereas CNB does not. Their calibration fits remain inside each base fitting subset but share globally fitted features. No proper-score or reliability diagnostic evaluates these probabilities [15]. LightGBM's stopping criterion uses the same validation fold later scored for macro-F1; log loss and macro-F1 are different objectives.

4.4. OOF Features and Selected Test Predictor

Let V_1, V_2, V_3 partition $I = \{1, \ldots, n\}$, and let $f(i)$ identify row i's fold. For base candidate b, let $h_b^{(-f)}$ be its probability-producing fit excluding V_f, with calibration or early stopping as specified. Training and test meta features differ:

$$\begin{aligned} z_i &= \left[h_b^{(-f(i))}(\phi(x_i))\right]_{b=1}^{6} \in \mathbb{R}^{18}, \\ \bar{z}(x) &= \left[\frac{1}{3}\sum_{f=1}^{3} h_b^{(-f)}(\phi(x))\right]_{b=1}^{6}. \end{aligned} \tag{3}$$

LightGBM additionally receives the numerical features, suppressed in this notation for readability. The notation excludes a fold from its base estimator fit, not from global preprocessing, class weights, or LightGBM stopping selection.

Three Meta-LR fits are trained from the single global OOF feature matrix using the same split scheme. Their test probabilities are averaged. The selected final predictor combines six averaged base outputs and the averaged Meta-LR output, so Meta-LR is an active component rather than merely an alternative candidate. If F_j is candidate j's pooled OOF macro-F1 and $p_j(x)$ its test probability vector, then

$$\alpha_j = \frac{F_j^2}{\sum_{k=1}^{7} F_k^2}, \qquad \hat{y}(x) = \arg\max_c \sum_{j=1}^{7} \alpha_j p_j(x)_c. \tag{4}$$

These weights use labeled development outcomes rather than a prespecified average. Figure 1 separates this prediction path from the independent profiling operation.

[1] Consulted API: https://scikit-learn.org/1.5/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html.

Table 2: Source settings (S) and realized widths printed by the updated notebook (R). Numerical features are appended only for LightGBM.

Branch	Input	Analyzer	Grams / document frequency	Cap	Width
Word	$s(x)$	Word; non-whitespace tokens	1–3; min 1, max 0.98	100,000	100,000
Character	$s(x)$	`char_wb`	2–5; min 2	80,000	80,000
Clean character	$C(x)$	`char`	1–4; min 2	50,000	50,000
Signal	$H(x)$	Word; non-whitespace tokens	1–2; min 1	10,000	105
Combined TF-IDF				240,000	230,105
LightGBM input	TF-IDF + numerical		20 fixed numerical features		230,125

Table 3: Invoked settings in the updated source (S). Unspecified settings depend on the unrecorded historical library versions.

Candidate	Input	Explicit configuration
LR_C8, LR_C3	TF-IDF	$C = 8$ or 3; saga; multinomial; max. 2,000 iterations; tolerance 10^{-3}; global class weights; seed 42.
SVC	TF-IDF	LinearSVC, $C = 0.5$; max. 3,000 iterations; global class weights; seed 42; isotonic calibration, CV = 2.
CNB	Two character branches	$\alpha = 0.05$; `norm=True`; isotonic calibration, CV = 2; no explicit class weights.
SGD	TF-IDF	Modified Huber; $\alpha = 10^{-4}$; max. 200 iterations; global class weights; seed 42; isotonic calibration, CV = 2.
LightGBM	TF-IDF + 20 numerical	Multiclass log loss; 63 leaves; rate 0.1; feature fraction 0.7; bagging fraction 0.8, frequency 5; min. child samples 20; $L_1 = 0.1$, $L_2 = 0.5$; `is_unbalance=True`; global sample weights; seed 42; max. 1,000 rounds; early-stopping patience 80.
Meta-LR	18 probabilities	$C = 10$; max. 500 iterations; multinomial; global class weights; seed 42; solver not overridden.

5. Recorded Results and Submission Provenance

5.1. Candidate and Ensemble Results

Table 4 preserves printed candidate scores and fold values. LR_C3 is the strongest base candidate. The weighted ensemble has pooled OOF macro-F1 0.5314, compared with 0.5306 for the simple average and 0.5221 for LR_C3 (R). The rounded weighted-versus-base difference is 0.0093, or 0.93 percentage points (A). It is descriptive; the source selects weights, stopping points, and candidates using evaluation information, and no untouched comparison or paired prediction file is supplied.

Pooled F1 is computed once from all OOF rows, not averaged over fold F1. OOF prediction uses a single held-out fit, whereas test prediction averages three fits at both levels. Assessing the latter requires a matched frozen procedure.

The threshold search evaluates $30 \times 30 = 900$ pairs for each of the two ensemble probability arrays, adjusting classes 0 and 1 while leaving class 2's divisor at one (S). Its search interval is 0.1 to 0.9, omitting the unadjusted pair $(1, 1)$. A lower best threshold score than the unadjusted score is therefore possible without a contradiction. The final source compares four variants and selects the unadjusted weighted ensemble. The displayed weighted thresholds 0.90, 0.87, and 1.00 are not on that selected prediction path.

5.2. Confusion Counts and Error Structure

Table 5 transcribes the nine integer annotations in the updated notebook's embedded confusion figure. Row sums match the raw CSV supports and total 47,817. Reconstruction yields macro-F1 0.5314177985 and accuracy 0.5803166238 (A), consistent with the printed 0.5314. The three classwise values agree with the figure's four-decimal labels.

Neutral has the lowest F1 and recall. Of 4,829 examples predicted Neutral, 2,583 have a toxic label; 3,652 actual Neutral examples are assigned a toxic label. The largest off-diagonal count is 7,502 Subtle examples predicted Explicit. Without paired row-level outputs, causes and joint candidate errors remain unknown. An always-Explicit predictor has arithmetic macro-F1 $2n_0/[3(n + n_0)] = 0.221275$ (A), not a measured baseline experiment.

5.3. Test Export and Platform Claims

The updated notebook prints 6,414 Explicit, 4,401 Subtle, and 1,140 Neutral test predictions (R). A newly supplied CSV contains exactly these class counts, 11,955 rows, columns `id,y_pred,parameters`, IDs matching the three-class test file in order, and constant `parameters`=0.183 (D). This establishes aggregate and schema consistency with the recorded export. Without fitted states or row-level historical probabilities, it does not authenticate the origin of every prediction.

The source writes additional files with declared NOP values 0.1, 0.3, and 1. Number of parameters (NOP) means a count under a stated model boundary, not a freely chosen denominator. A zero-valued loop entry is skipped because it equals the initial profiled value; a later summary recommends that filename despite no matching save. We dis-

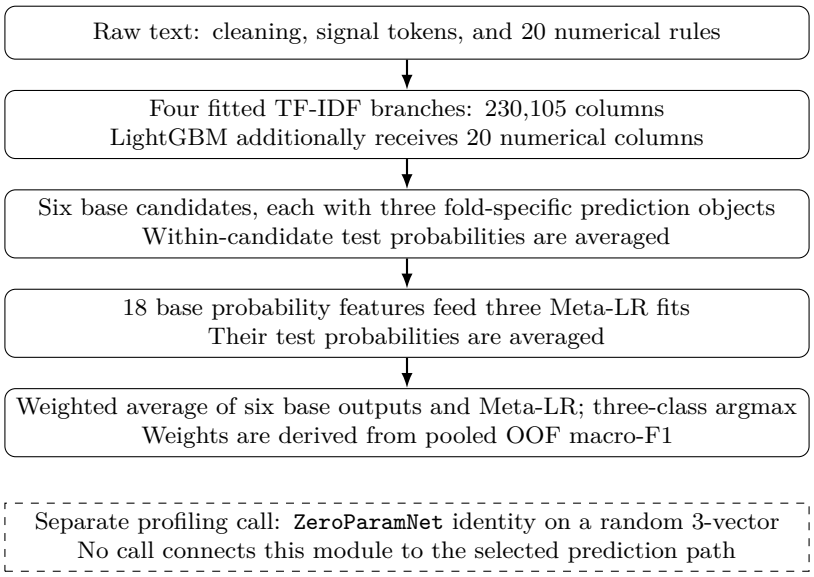


Figure 1: Source-derived computation (S), not a measured deployment benchmark. Base outputs also enter the final weighted average directly, as specified in Equation 4.

Table 4: Recorded pooled and fold macro-F1 (R). Ensemble variants are scored on the same pooled OOF rows; no uncertainty interval is inferred from dependent folds.

Candidate	Role	Pooled	Fold 1	Fold 2	Fold 3
LR_C8	Base	0.5169	0.5158	0.5122	0.5227
LR_C3	Base	0.5221	0.5209	0.5171	0.5281
SVC	Base	0.4824	0.4860	0.4827	0.4782
CNB	Base	0.4391	0.4403	0.4356	0.4414
SGD	Base	0.4691	0.4675	0.4719	0.4678
LightGBM	Base	0.5160	0.5154	0.5136	0.5191
Meta-LR	Stacker / ensemble member	0.5218	0.5183	0.5169	0.5299
Weighted	Selected ensemble	0.5314	–	–	–
Simple average	Alternative	0.5306	–	–	–
Weighted + thresholds	Alternative	0.5300	–	–	–
Average + thresholds	Alternative	0.5287	–	–	–

tinguish these source operations from platform acceptance. The new author-reported public/private scores remain X evidence.

5.4. Timing and Retained State

The recorded LR training sections take 2,601.7 and 2,625.0 seconds, and LightGBM takes 3,802.8 seconds (R). These three sections alone total 9,029.5 seconds, about 150.5 minutes (A), before other work. The notebook metadata reports about 152.4 minutes. Thus its concluding "25–35 minutes" claim is inconsistent with its retained timings. Those timings are historical output, not a newly controlled runtime benchmark, and no inference latency measurement is supplied.

LightGBM reports best iterations 149, 173, and 182 (R). The inspected source serializes no fitted vectorizers, base models, boosters, meta estimators, or historical OOF arrays. In-memory test arrays are generated before export, but the absence of saved model files prevents fresh-process raw-text replay. Historical environment uncertainty remains even though the source and recorded dimensions can now be inspected.

6. Evaluation Validity

6.1. Different Types of Evaluation Exposure

The retained score describes development predictions. Table 6 separates mechanisms by the information they use and their structural reach. The tiers prioritize validation work; they are not measured effect sizes or a universal ordering of harm.

The TF-IDF fits precede fold construction, but test text is only transformed. Thus this is intra-training covariate exposure, not fitting on the unlabeled test set. Global class weights use $w_{I,c} = |I|/(3n_{I,c})$. Regenerated fold-local weights differ from these global values by at most 0.005506% relatively (A), reflecting the close stratified class counts.

Table 5: Selected weighted-ensemble OOF confusion counts (R), with classwise metrics reconstructed from the displayed counts (A). Rows are actual labels; columns predicted labels.

Actual	Explicit	Subtle	Neutral	Support	Precision	Recall	F1
Explicit	16,464	6,331	961	23,756	0.6501	0.6930	0.6709
Subtle	7,502	9,039	1,622	18,163	0.5118	0.4977	0.5046
Neutral	1,360	2,292	2,246	5,898	0.4651	0.3808	0.4188
Predicted	25,326	17,662	4,829	47,817	Macro-F1 0.531418 (A)		

Table 6: Structural exposure taxonomy (S/A/P). Magnitude is unmeasured unless an observed diagnostic is stated; an observed score gap is not a bias bound.

Mechanism / tier	Information and reach	Interpretation and required control
TF-IDF / covariate exposure	Label-free validation text enters vocabulary, frequency filters, caps and IDF.	Transductive representation exposure. Fit each branch on the permitted training subset; compare with the global arm. No label-path claim.
Class weights / aggregate exposure	Three class counts include validation labels; fitting sample weights inherit them.	Low-dimensional label statistics. Here stratification makes fold-local values differ by at most 0.005506% relatively (A); measure any prediction effect separately.
Early stopping / supervised selection	LightGBM assessment labels select boosting rounds through log loss.	Reuse of assessment outcomes. Use a stopping partition inside the training subset, then assess the frozen fit.
Meta-stacking / supervised feature ancestry	Meta-training features are generated by base fits containing meta-assessment labels.	Full supervised dependency path. Rebuild inner OOF features within every outer training subset.
Ensemble weights / aggregate selection	Seven pooled F1 scores determine normalized squared weights.	Near-uniform weights; observed gain 0.0008. Estimate optimism only with protected assessment labels.
Threshold and candidate choice / repeated selection	Two grids of 900 choices and four final variants use the same labeled OOF rows.	Multiple adaptive comparisons. Freeze decisions within training-only selection partitions.

This small statistic difference should not be presented as equivalent in scope to the meta learner's full supervised ancestry. It does not by itself prove a negligible prediction effect.

6.2. Nonnested Meta-Training Dependencies

For a meta learner $q^{(-a)}$ trained on $\{(z_i, y_i) : i \notin V_a\}$ and evaluated on V_a, take a meta-training row $i \in V_b$, $b \neq a$. Its features came from base fits on $I \setminus V_b$, a set containing V_a. Thus

$$\begin{aligned} y_{V_a} &\longrightarrow h_m^{(-b)} \longrightarrow z_i \\ &\longrightarrow \text{meta-training data} \longrightarrow q^{(-a)}. \end{aligned} \tag{5}$$

A row in V_a does receive base features from an estimator fit excluding V_a. However, the meta model evaluating that row was trained on other features whose construction used y_{V_a}. Row-level OOF exclusion therefore does not isolate the complete stacker. This structural dependency remains even if global preprocessing and class weights are provisionally accepted.

Rebuilding base OOF features only once for all meta folds preserves this path. It must instead be regenerated within the training partition of each outer assessment split, and again within the allowed fitting subset of each selection split. Equation 5 establishes missing isolation, not the magnitude or direction of score distortion. The nested comparison in the supplement is prospective; its result remains an explicit author task.

6.3. Stopping, Near-Uniform Weights, and Selection

LightGBM uses its assessment labels for stopping and later reports macro-F1 on the same rows. Reporting a different objective does not create a new assessment sample. Ensemble weights are functions of the pooled OOF F1 values, so each evaluation label can also affect the aggregation rule.

Using the seven printed four-decimal candidate scores in Equation 4 gives approximate weights 0.154991, 0.158125, 0.134992, 0.111846, 0.127651, 0.154452, and 0.157943 for LR_C8, LR_C3, SVC, CNB, SGD, LightGBM, and Meta-LR respectively (A). Uniform weight is $1/7 = 0.142857$; the largest absolute departure is about 0.031011. The source used unrounded scores, so these are reconstructed approximations, not recovered fitted weights.

The recorded weighted score 0.5314 exceeds the simple average 0.5306 by 0.0008, or 0.08 percentage points (A). This is a small observed development-set gain. It is *not*

an upper bound on leakage or selection optimism: both probability arrays share other evaluation dependencies, and no untouched paired comparison is available. We therefore give this mechanism narrower emphasis than meta-feature ancestry.

Each threshold grid assesses 900 choices on the same OOF labels. Their selected results and unadjusted alternatives are then compared to choose one of four final variants. These are selection opportunities [3, 4]; no measured bias is assigned to them or to the rounded 0.0093 ensemble-versus-base difference.

6.4. Dependence and Uncertainty

The three fitting subsets overlap: each pair shares roughly one-third of the full corpus, or half of either fitting subset. They are not independent repetitions. Stratification addresses class balance but does not protect against duplicates, shared authors, threads, or temporal dependence. Seed-42 fold indices are now regenerated from the supplied CSV using scikit-learn 1.8.0. They are saved with IDs and labels. They are a new reconstruction, not an authenticated historical fold file; their duplicate counts appear below.

There is no universally unbiased K-fold variance estimator based only on observed fold errors [16]. Paired comparisons also require retained row level predictions and a procedure appropriate to the data's independent units [17]. We report neither a naive fold-based interval nor a significance claim. Such an interval would not resolve the fitting and selection exposures already traced.

6.5. New Checks and Controlled Comparisons

We regenerate `StratifiedKFold(3, shuffle=True, random_state=42)` without changing row order. A repeated-text group means two or more rows with exactly the same raw string. Of 461 such training groups containing 1,082 rows, 308 groups containing 771 rows span multiple regenerated folds (D). Conflicting labels occur in 52 repeated-text groups containing 130 rows; 36 of those groups span folds. These counts reveal split and annotation dependence, not its effect on macro-F1. No normalization or near-duplicate criterion is used.

The LR script fits all four branches globally in one arm and only on each training fold in the other. Both retain the original global class weights to isolate TF-IDF scope. They use $C = 3$, SAGA, multinomial loss, tolerance 10^{-3}, maximum 2,000 iterations, and seed 42. Software versions, warnings, iterations, probabilities and fold IDs are saved. All six LR fits reach the 2,000-iteration cap and emit convergence warnings (new R). Thus the result describes the source training budget, not converged optima. The global score rounds to the historical 0.5221, but does not authenticate that unknown environment. A fully corrected arm should additionally recompute class weights within each training subset.

The complete notebook is not claimed impossible to rerun. Its historical outputs cannot be replayed from saved fitted objects because those objects and versions were not retained. This audit environment has an 8 GiB memory limit, CPU execution and scikit-learn 1.8.0. LightGBM 4.7.0 was made available for a new run; PyTorch profiling is separate from the classical predictor. The full-data paired LR comparison is completed here. The 54 nested base-candidate objects, stopping refits and matched non-nested arm exceed this revision's training budget; the supplemental protocol leaves its result as [**TODO-AUTHOR: execute both stacking arms and insert paired macro-F1 results**]. A new training run is replication evidence, not recovery of the original fitted states.

7. Parameter Boundaries and Score Provenance

7.1. The Identity Module Is a Different Object

The source defines `ZeroParamNet.forward(x)` to return x unchanged. It has no trainable parameters. Profiling, when available, calls this module on a random three-element vector after the selected predictions already exist (S). No invocation applies the module to raw text, base probabilities, meta features, or selected outputs. Its zero count therefore describes a separate identity operation, not the harmful-content predictor.

The retained output prints `Manual NOP: 0` and `Final thop NOP: 0`, without the success line printed by the successful profiling branch. The variable is initialized to zero and remains zero when THOP is unavailable or fails. Consequently, the final printed zero alone cannot establish successful profiling. The source uses one positional input to this separate module, whereas the supplied competition description illustrates a two-input model call with token IDs and an attention mask. These interfaces do not determine a whole-system accounting boundary.

7.2. Structural Inventory of the Selected Predictor

The selected path needs six three-fold base candidates, three Meta-LR fits, seven weights, four fitted vocabularies/IDF arrays and fixed text/numerical rules. No learned PyTorch head is invoked.

At recorded TF-IDF width $D = 230{,}105$, each three-class multinomial LR fit has $3D$ coefficients and three intercepts. The two LR candidates each require three fits for their stored test averages. Meta-LR has 18 input probabilities and three output scores:

$$\begin{aligned} N_{\text{one base LR}} &= 3D + 3 = 690{,}318, \\ N_{\text{six base LR}} &= 6(3D + 3) = 4{,}141{,}908, \\ N_{\text{three Meta LR}} &= 3(3 \times 18 + 3) = 171. \end{aligned} \tag{6}$$

These are arithmetic scalar inventories for specified matrices (A), not measured memory occupancy or a complete contest NOP. Coefficients for SVC and SGD, CNB state, calibrators, LightGBM trees, vocabularies, and IDF arrays

Table 7: New evidence and pending comparisons. D/A results are checks or calculations; new LR scores are explicitly distinguished from historical R. No missing result is inferred from the 0.5314 record.

Check / comparison	Result	Evidence and interpretation
LR_C3, global TF-IDF	New pooled F1 0.522061	New R; same full CSV and folds as the paired arm.
LR_C3, training-fold TF-IDF	New pooled F1 0.518872	New R; fixed global class weights and source solver/settings.
Global minus fold-fitted LR	0.003189 macro-F1	A; difference in this paired run, not a general bias bound.
Non-nested / nested Meta-LR	Historical 0.5218 / **[TODO-AUTHOR: run nested stack]**	R/P; compare newly executed matched arms, not one historical score against one new environment.
Exact-text groups spanning folds	308 groups; 771 rows	D; regenerated seed-42 indices.
Repeated-text groups with label conflict	52 groups; 130 rows	D; 36 groups also span folds.
Squared-F1 weight range	Approx. 0.111846–0.158125	A; computed from rounded scores.
Weighted minus uniform ensemble	0.0008 macro-F1	A/R; observed development difference, not a bias bound.
Updated CSV consistency	11,955 rows; NOP field 0.183	D; IDs/order and class counts match the recorded export.

remain outside this subtotal. Internal calibration can retain multiple fitted classifiers/functions per outer object; historical defaults and fitted internals are unknown [22].

No complete serialized artifact, peak RAM measurement, raw-text inference latency, or energy benchmark is available. Scalar counts, fitted state, bytes, RAM, latency and energy are distinct [18]. Comparisons need the same boundary and models that generate probabilities for unseen text.

7.3. Fractional Export Values Are Not Measurements

For the public-datathon record, the source sets the main export field by rounding $F/2.9$ to three decimal places (S), producing 0.183 (R/D). This is a target-derived denominator, not a structural parameter measurement. The assigned values 0.183, 0.1, and 0.3 are fractional, whereas a count of scalar parameters is an integer. The separate identity operation has zero trainable parameters; the selected fitted ensemble has additional state. These observations describe implementation and measurement provenance, not a verified organizer ruling or a claim about author intent.

With $F = 0.5314177984731835$ and $\epsilon = 5 \times 10^{-16}$, the local ratio $F/(N+\epsilon)$ gives about 1.0628×10^{15} at $N = 0$ and 2.9039 at $N = 0.183$ (A). These are arithmetic projections, not the new author-reported public/private platform scores. Changing the case name does not change the technical facts or remove the need to document the applicable accounting rules.

7.4. Authoritative Scoring Remains Unresolved

The supplied details document gives $0.4F + 0.6/[\log(N + 1) + \epsilon]$ without a log base, but later quotes $F/(N + \epsilon)$. The original formula image, official scoring code, and classical-model count boundary remain unavailable. The notebook's local ratio is source evidence, not verification of the contest contract. Public and private platform values 3.78469 and 3.78930 are author reports (X); hidden-set macro-F1, uploaded-file identity, and acceptance are not inferred from them. Earlier scores from other versions are confined to Supplement S1.

[TODO-AUTHOR: provide the scorer and count rule, plus a receipt linked to the submitted file hash]. Both naming versions retain these limitations. The anonymized version describes the case as a public datathon; it does not conceal a changed accounting calculation.

8. Corrected Prospective Protocol

Freeze data bytes, label meanings, sampling units and permitted uses before assessment. Inspect duplicate/conflict groups; grouping protects repeated text while label conflicts require documented treatment. Keep the regenerated row-stratified comparison separate from a future group-aware evaluation.

For each outer training subset T_a and assessment subset E_a, build new inner three-fold base OOF features using only T_a. Fit the four TF-IDF branches and class weights within permitted fitting subsets. Calibration remains inside each base fitting subset. LightGBM stopping uses a further training-only partition, never E_a. Fit Meta-LR on the new inner OOF features, and generate E_a base features by averaging the same inner fits. This evaluates a specified fold-averaged base predictor. It does not silently equate it with a different final deployment ensemble.

Run the original non-nested arm in the same new environment and save its paired probabilities. The fully corrected nested arm changes preprocessing, class statistics, stopping and feature ancestry jointly; its score difference is a protocol comparison, not a causal estimate of stacking alone. A separate nesting-only sensitivity study must hold the other operations fixed. Reserve ensemble weights, threshold search and candidate choice for training-only selection if evaluating the full selected ensemble. The requested Meta-LR comparison does not automatically validate that full ensemble.

Preserve IDs, labels, folds, probabilities, versions, convergence warnings, fitted objects and stopping choices. Serialize/reload the complete raw-text predictor and check probability equivalence before measuring bytes, peak RAM, latency or energy [19]. Define contest NOP using its original rules. Supplement S2 supplies the runnable LR script and complete stacking protocol; unperformed studies retain explicit author tasks.

9. Limitations and Responsible Interpretation

The updated source and retained outputs are inspectable, and the supplied CSVs agree with its printed supports. However, no hash recorded during the historical run establishes that those exact bytes were used. The audit regenerates folds, checks a new export, and executes a controlled LR comparison. It does not recover historical OOF rows, inspect historical fitted states, or verify platform acceptance. Unknown package versions and contradictory device metadata limit replay. Exact-string duplication, conflicting labels and regenerated cross-fold allocation are checked. Semantic duplicates and shared-user/thread structure remain unexamined.

The threats are distinct: internal, because fitting and selection use evaluation information; external, because one record does not establish a population claim; construct-related, because label definitions lack a complete annotation manual; statistical, because folds and selection outcomes are dependent; and measurement-related, because the counted identity object differs from the selected predictor. Neither the direction of score distortion nor deployment cost is measured.

The three-way argmax is a research prediction, not a moderation policy. Substring matching can activate inside benign words, and identity-related cues do not model whether a speaker endorses, condemns, quotes, or reclaims an expression [8, 7]. No operating threshold, contextual safety, appeal process, or deployment benefit has been established. No legal basis, consent, ethics approval, or independent redistribution permission is inferred from public-source collection or from a supplied license description.

10. Conclusion

This case study reconstructs the updated ensemble record: 47,817 training rows, four fitted TF-IDF branches, six base candidates, a Meta-LR component, and a selected weighted ensemble. Its reconstructed macro-F1 0.531418 agrees with the notebook's 0.5314; the rounded comparison with LR_C3 at 0.5221 remains descriptive. Regenerated folds reveal repeated and conflicting text groups; the new CSV matches recorded export structure and class totals. Historical versions remain supplemental.

Three findings transfer. Row-level OOF exclusion does not isolate a full procedure when evaluation labels remain ancestors of meta-training features, stopping choices, or combination weights. A zero-parameter identity operation does not account for the fitted state that produces harmful-content predictions. A fractional denominator selected to achieve a desired score is not a parameter measurement. Every predictive or efficiency claim should name its inference object, fitting and selection ancestry, accounting boundary, and supporting evidence. This remains an inspectable methodological case, not a validated deployment system.

Data and Code Availability.

The review artifact contains source, CSV checks, fold indices, scripts, outputs and a digest ledger. [**TODO-AUTHOR: insert an anonymous review-artifact URL and permanent dataset citation permitted by its terms**]. [**TODO-AUTHOR: insert the approved public code/archive URL after blind review**]. The publicly named repository is documented in the author revision report; it is not substituted for an anonymous review link. Release availability does not itself establish redistribution permission.

Author Declarations.

[**TODO-AUTHOR: state actual funding or an author-confirmed no-funding declaration**]. [**TODO-AUTHOR: state competing interests**]. [**TODO-AUTHOR: state documented ethics approval/exemption, collection basis, data-use and redistribution permissions**]. [**TODO-AUTHOR: disclose the prior manuscript's venue/status, overlap and any simultaneous consideration**]. AI assistance was used for literature organization, source inspection, arithmetic/data checks, drafting and typesetting. [**TODO-AUTHOR: all authors approve this disclosure and the final text; document any additional AI use**].

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